Aakash Tripathi, Ph.D.

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Current Research and Development Focus

1. Multimodal Foundation Models for Digital Biology - MINDS and HONeYBEE Frameworks: Leading development of open-source multimodal AI platforms: HONeYBEE (npj Digital Medicine, IF: 15.1) for unified cancer data embeddings (18+ GitHub stars, 400+ PyPI downloads) and MINDS for scalable data aggregation (7+ GitHub stars, 2000+ PyPI downloads). Created HuggingFace dataset with 3,500+ downloads enabling community research.
2. Large-Scale Training and Inference for Biomedical AI: Adapting Transformers and Graph Neural Networks for processing thousands of cancer cases, accelerating drug discovery pipelines and identifying novel therapeutic targets through pattern recognition across multi-omics data at scale.
3. Self-Normalizing Multi-Omics Platform for Precision Medicine – SeNMo: Created SeNMo, a deep learning system achieving 85% accuracy in survival prediction and treatment response by integrating incompatible molecular data formats, enabling personalized treatment recommendations across diverse molecular assays.
4. AI-Powered Medical Image Analysis and Digital Pathology; Building automated systems for gigapixel pathology and radiological analysis with 95% diagnostic accuracy, reducing pathologist workload by 60% while maintaining clinical-grade performance across multiple cancer types.
5. Clinical NLP with Multi-Agent AI Systems – CLEVER: Developing CLEVER, a consensus-based multi-agent system achieving 92% accuracy in extracting structured data from unstructured pathology reports, transforming years of clinical text into research-ready databases for accelerated discovery.

Employment

H. Lee Moffitt Cancer Center and Research Institute — Sept 2025 – Present
Tampa, FL, USA
Machine Learning Engineer I

- Department of Machine Learning
- Paper published in Springer Nature npj Digital Medicine
- Developed open-source multimodal processing framework called HoneyBee with 18+ stars and 400+ downloads on PyPi.
- Made a multimodal dataset publicly available on HuggingFace using HoneyBee with over 3,500 downloads and 11 likes.
- Developed an agentic document extraction tool that provides grounded and an auditable system for extraction key clinical variables, patient timelines, and ICD billing codes, called CLEVER.
- Deploying CLEVER at an institution scale to process all medical records and provide a source of document quality control.

University of South Florida Sept 2022 – Aug 2025
Tampa, FL, USA
Graduate Research Assistant

- Department of Machine Learning
- Supervisors: Dr. Yasin Yilmaz and Dr. Ghulam Rasool
- Developed an open-source data aggregation solution called MINDS with 7+ GitHub stars and over 2000 downloads on PyPi.

Published papers in MDPI Sensors, Frontiers in Artificial Intelligence, International journal of molecular sciences, and Cancer Informatics.

Education

University of South Florida Aug 2022 – Aug 2025
Tampa, FL, USA
Ph.D., Electrical Engineering

- PhD Thesis: PhD Thesis: Tripathi, A. G. (2025). *Embedding-Based Deep Learning Frameworks for Multimodal Oncology Data Integration* (Doctoral dissertation, University of South Florida).
- Department: Department of Electrical Engineering
- Supervisors: Dr. Yasin Yilmaz and Dr. Ghulam Rasool

Rowan University Sep 2018 – May 2022
Glassboro, NJ, USA
B.S., Electrical and Computer Engineering

- Published papers in IEEE International Symposium on Circuits and Systems (ISCAS) and Circuits, Systems, and Signal Processing.
- Provisional patents on sEMG sensors for AI/VR controls.

Teaching Experience

Teaching Assistant: Mayo Clinic AI Summit Workshop on "Scalable Multimodal AI in Oncology Using HONeYBEE: From Embeddings to Clinical Impact". https://ai-summit.com/ai-summit/home July 8, 2025

Teaching Assistant: Building Transformer-based Natural Language Processing, NVIDIA Deep Learning Institute, Public Student Workshop, North America, Virtual. https://www.nvidia.com/en-us/training/instructor-led-workshops/natural-language-processing/ February 23, 2024

Teaching Assistant: Building Transformer-based Natural Language Processing, NVIDIA Deep Learning Institute, Public Student Workshop, North America, Virtual. https://www.nvidia.com/en-us/training/instructor-led-workshops/natural-language-processing/ June 29, 2022

Teaching Assistant: Fundamentals of Deep Learning for Computer February 22, 2020

Vision, NVIDIA Sponsored Workshop, Rutgers Business School, Rutgers University, NJ, USA. https://www.nvidia.com/en-us/training/instructor-led-workshops/fundamentals-of-deep-learning/

Publications

Peer-Reviewed Publications

[1] **Aakash Tripathi,** Waqas, A., Schabath, M. B., Yilmaz, Y., & Rasool, G. (2025). HONeYBEE: enabling scalable multimodal AI in oncology through foundation model-driven embeddings. *npj Digital Medicine*, *8*(1), 622. https://doi.org/10.1038/s41746-025-02003-4

[2] Flack, D., **Tripathi, A.,** Waqas, A., Rasool, G., & Dera, D. (2025). Robust Multimodal Fusion for Survival Prediction in Cancer Patients. *Cancer Informatics*, *24*, 11769351251376192. https://doi.org/10.1177/11769351251376192

[3] Waqas, A., **Tripathi, A**., Ahmed, S., Mukund, A., Farooq, H., Johnson, J. O., ... & Rasool, G. (2025). Self-normalizing multi-omics neural network for pan-cancer prognostication. *International journal of molecular sciences*, *26*(15), 7358. https://doi.org/10.3390/ijms26157358

[4] Waqas, A., **Tripathi, A.**, Ramachandran, R. P., Stewart, P. A., & Rasool, G. (2024). Multimodal data integration for oncology in the era of deep neural networks: a review. *Frontiers in Artificial Intelligence*, *7*, 1408843. https://doi.org/10.3389/frai.2024.1408843

[5] **Tripathi, A.,** Waqas, A., Venkatesan, K., Yilmaz, Y., & Rasool, G. (2024). Building flexible, scalable, and machine learning-ready multimodal oncology datasets. *Sensors*, *24*(5), 1634. https://doi.org/10.3390/s24051634

[6] Ahmed, S., Nielsen, I. E., **Tripathi, A.**, Siddiqui, S., Ramachandran, R. P., & Rasool, G. (2023). Transformers in time-series analysis: A tutorial. *Circuits, Systems, and Signal Processing*, *42*(12), 7433-7466. https://doi.org/10.1007/s00034-023-02454-8

[7] Epifano, J. R., Silvestri, A., Yu, A., Ramachandran, R. P., **Tripathi, A**., & Rasool, G. (2023, May). A comparison of feature selection techniques for first-day mortality prediction in the icu. In *2023 IEEE International Symposium on Circuits and Systems (ISCAS)* (pp. 1-5). IEEE. https://doi.org/10.1109/ISCAS46773.2023.10182228

Dissertation

[1] Tripathi, A. G. (2025). *Embedding-Based Deep Learning Frameworks for Multimodal Oncology Data Integration* (Doctoral dissertation, University of South Florida).

Pre-print / Under-Review Publications

[1] **Tripathi, A.**, Nielsen, I. E., Umer, M., Ramachandran, R. P., & Rasool, G. (2025). Explainable AI in Genomics: Transcription Factor Binding Site Prediction with Mixture of Experts. *ArXiv*, arXiv-2507.

[2] **<u>Tripathi, A.</u>**, Waqas, A., Schabath, M. B., Yilmaz, Y., & Rasool, G. (2025). EAGLE: Efficient Alignment of Generalized Latent Embeddings for Multimodal Survival Prediction with Interpretable Attribution Analysis. *arXiv preprint arXiv:2506.22446*.

[3] Garcia-Fernandez, C., Felipe, L., Shotande, M., Zitu, M., **<u>Tripathi, A.</u>**, Rasool, G., ... & Valdes, G. (2025). Trustworthy AI for Medicine: Continuous Hallucination Detection and Elimination with CHECK. *arXiv preprint arXiv:2506.11129*.

[4] **<u>Tripathi, A.</u>**, Waqas, A., Venkatesan, K., Ullah, E., Khan, A., Khalil, F., ... & Rasool, G. (2025). Employing Consensus-Based Reasoning with Locally Deployed LLMs for Enabling Structured Data Extraction from Surgical Pathology Reports. *medRxiv*, 2025-04.

[5] Felipe, L., Garcia, C., Naqa, I. E., Shotande, M., **<u>Tripathi, A</u>**., Rudrapatna, V., ... & Valdes, G. (2025). TheBlueScrubs-v1, a comprehensive curated medical dataset derived from the internet. *arXiv preprint arXiv:2504.02874*.

[6] Waqas, A., **<u>Tripathi, A.</u>**, Stewart, P., Naeini, M., Schabath, M. B., & Rasool, G. (2024). Embedding-based multimodal learning on pan-squamous cell carcinomas for improved survival outcomes. *arXiv preprint arXiv:2406.08521*.

Talks, Presentations, & Abstracts

[1] Waqas A., Tripathi, A., Naeini, M., Stewart, P., Schabath, M. B., & Rasool, G. (2025). Using Patient Embeddings From Foundation Models for Enhanced Survival Analysis in Lung Squamous Cell Carcinoma. American Journal of Respiratory and Critical Care Medicine, 211(Abstracts), A3108-A3108.

[2] Waqas A., Tripathi, A., Naeini, M., Stewart, P., Schabath, M. B., & Rasool, G. (2025). PARADIGM: an embeddings-based multimodal learning framework with foundation models and graph neural networks. Cancer Research, 85(8_Supplement_1), 991-991.

[3] Tripathi, A. G., Waqas A., Yilmaz, Y., Schabath, M. B., & Rasool, G. (2025). Predicting treatment outcomes using cross-modality correlations in multimodal oncology data. Cancer Research, 85(8_Supplement_1), 3641-3641.

[4] Tripathi, Aakash, Asim Waqas, Kavya Venkatesan, Ehsan Ullah, Marilyn Bui, and Ghulam Rasool. "1391 AI-Driven Extraction of Key Clinical Data from Pathology Reports to Enhance Cancer Registries." Laboratory Investigation 105, no. 3 (2025). https://doi.org/10.1016/j.labinv.2024.103629.

[5] Aakash Tripathi, Asim Waqas, Kavya Venkatesan, Ehsan Ullah, Matthew B. Schabath, Marilyn Bui, Ghulam Rasool, "AI-Driven Extraction of Key Clinical Data from Pathology Reports to Enhance Cancer Registries", Presentation at the United States and Canadian Academy of Pathology (USCAP) 114th Annual Meeting, March 22-27, 2025, Boston, Massachusetts, 2025.

[6] Ghulam Rasool, Aakash Tripathi, Asim Waqas, Ehsan Ullah, Marilyn Bui, "Extraction of Discrete Information from Pathology Reports Using Local and Private LLMs", Oral Presentation at the Digital Pathology Association, Pathology Visions 2024.

[7] Asim Waqas, Aakash Tripathi, Ashwin Mukund, Paul Stewart, Mia Naeini, Ghulam Rasool, "BIO24-031: Hierarchical Multimodal Learning on Pan-Squamous Cell Carcinomas for Improved Survival Outcomes" Journal of the National Comprehensive Cancer Network, 22(2.5), April 2024. https://doi.org/10.6004/jnccn.2023.7137.

[8] Asim Waqas, Aakash Tripathi, Sabeen Ahmed, Ashwin Mukund, Paul Stewart, Mia Naeini, Hamza Farooq, Ghulam Rasool. "SeNMo: A self-normalizing deep learning model for enhanced multi-omics data analysis in oncology," American Association for Cancer Research 84.6_Supplement (2024): 908-908. https://doi.org/10.1158/1538-7445.AM2024-908

[9] Aakash Tripathi, Asim Waqas, Ghulam Rasool. "Unifying Multimodal Data, Time Series Analytics, and Contextual Medical Memory: Introducing MINDS as an Oncology-Centric Cloud-Based Platform." National Comprehensive Cancer Network, April 2024. https://doi.org/10.6004/jnccn.2023.7305

[10] Aakash Tripathi, Asim Waqas, Yasin Yilmaz, Ghulam Rasool. "Multimodal Transformer Model Improves Survival Prediction in Lung Cancer Compared to Unimodal Approaches," American Association for Cancer Research 84.6_Supplement (2024): 4905-4905. https://doi.org/10.1158/1538-7445.AM2024-4905

[11] Asim Waqas, Aakash Tripathi, Ashwin Mukund, Paul Stewart, Mia Naeini, Ghulam Rasool. "Pan-cancer Learning for Survival Prediction." USF AI+X Symposium, 29 September 2023.

[12] Aakash Tripathi, Asim Waqas, Yasin Yilmaz, Ghulam Rasool. "Advancing Cancer Research Through Integrated Multimodal Data Analysis: A Comprehensive Framework for Precision Oncology" USF AI+X Symposium, 29 September 2023.

[13] Aakash Tripathi, Asim Waqas, Yasin Yilmaz, Ghulam Rasool. "HoneyBee: Developing Multimodal AI-Ready Datasets from Public Cancer Repositories" Nvidia GTC 2025, March 20, 2025. https://www.nvidia.com/gtc/session-catalog/?tab.catalogallsessionstab=16566177511100015Kus&search=P74176&ncid=em-even-124008-vt33-23spring#/session/1734302448222001uamP

[14] Aakash Tripathi, Asim Waqas, Ghulam Rasool. "CLeVER Multi-Agent AI Orchestration for Temporal-Aware Clinical Variable Extraction from Unstructured Medical Records" Dr. Robert Gillies Machine Learning Workshop in Cancer, October 30, 2025.

Peer Reviewing

- Springer Nature Scientific Reports — 2025
- Springer Nature npj Digital Medicine — 2025
- Springer Nature Discover Artificial Intelligence — 2025
- Springer Nature Journal of Medical Systems — 2025
- Springer Nature BMC Medical Informatics and Decision Making — 2025
- Springer Nature Discover Applied Sciences — 2025

2025

Technical Skills

- **Programming Languages**: Python, MATLAB/Octave, JavaScript
- **Technologies/Frameworks**: PyTorch, AWS, REACT, SQL & NoSQL Databases, Docker & Podman, SLURM, HuggingFace, vLLM, Ollama, LangChain, Deep Agents, Nvidia RAPIDS, HPC, UNIX shell, bash scripting

Honors, Awards, & Professional Memberships

- Named on multiple funded and applied research grants and funding from Federal/State agencies.
- 2nd position in the Moffitt Cancer Center Annual 2024 Bio-Data Club Hackathon – project "Patient Data Vectors – An Efficient Cancer Research Framework" 12-13 December 2024.
- Best poster award in the 2024 Gillies Dr. Robert Gillies Machine Learning Workshop in Cancer – "HONeYBEE: Enabling Scalable Multimodal AI in Oncology Through Foundation Model–Driven Embeddings", award price $1,000
- University of South Florida, Graduate Assistantship Award, 2022 – 2025.
- Dean's List for academic excellence, Sept 2018 – May 2022.
- IEEE Eta Kappa Nu (IEEE-HKN) Honors Society Member, April 16, 2024 – present
- Institute of Electrical and Electronics Engineers (IEEE) Member January 2019 – present

Mentoring & Training Experience

Collaborator and Co-Investigator with Mr. Siddharth Sivaram, Ph.D. student, Department of Electrical Engineering, University of South Florida, Tampa, FL, USA.	Aug 2025 – present
Collaborator and Co-Investigator with Ms. Hanieh Ajami, Ph.D. student, Department of Computer Science, University of South Florida, Tampa, FL, USA.	Jan 2025 – present
Collaborator and Co-Investigator with Mr. Nikolas Koutsoubis, Ph.D. student, Department of Machine Learning, Moffitt Cancer Center and Research Institute, Tampa, FL, USA.	Aug 2024 – present
Collaborator with Mr. Dominic Flack, Graduate student, Chester F. Carlson Center for Imaging Science, College of Science, Rochester Institute of Technology, Rochester, NY, USA.	Aug 2024 – Feb 2025
Collaborator and Co-Investigator with Ms. Kavya Venkatesan, Research Intern, Department of Machine Learning, Moffitt Cancer Center and Research Institute, Tampa, FL, USA.	May 2023 – present
Collaborator and Co-Investigator with Mr. Asim Waqas, Ph.D. student, Department of Machine Learning, Moffitt Cancer Center and Research Institute, Tampa, FL, USA.	Aug 2022 – Aug 2025